

Fundamentals of Signals and Systems (EE203)

Course Content:

UNIT 1: Introduction to Signals

Definition and classification of signals, Basic signal operations, Elementary signals (impulse, step, ramp, exponential, and sinusoidal). (3 Lectures)

UNIT 2: Introduction to Systems

Definition and classification of systems, LTI systems, properties of LTI systems, and Convolution. (5 Lectures)

UNIT 3: Fourier Series and Fourier Transform

Fourier Series, Continuous-Time Fourier Transform (CTFT), and applications in signal analysis (8 Lectures)

UNIT 4: Laplace and Z-Transform

Laplace Transform and Application to LTI System Analysis, Z-Transform and Application in discrete-time LTI systems (8 Lectures)

UNIT 5: Discrete Fourier Transform (DFT)

Sampling, Discrete Fourier Transform (DFT), Properties of DFT (8 Lectures)

UNIT 6: Frequency Response and Filter Design

Frequency response of LTI systems, System function, poles, and zeros, Stability analysis, analog and digital filters (8 Lectures)